SWRT for nonlinear four-wave interactions in ecWAM: installation and quick guide

SWRT (sparse Webb–Resio–Tracy) computes the nonlinear four-wave interactions S_{nl} from K configurations of the WRT quadrature table, each a quadruplet and its mirror image, with one optimized coefficient per configuration. This patch adds SWRT to ecWAM as an alternative to the Discrete Interaction Approximation (DIA) in `SNONLIN`. In a 24-h global test, SWRT keeps the significant wave height within 1 % of WRT at 1.6 times the time per step of DIA.

The changes, the SWRT files and this guide are on the branch `swrt` of Jialunx/ecwam, a fork of ecmwf-ifs/ecwam at commit `e1779dc0` (ecWAM 1.5.14).

1. Changes to ecWAM

The patch adds one source file and changes eight others. The default `INLSCHEME = 0` leaves ecWAM unchanged.

File	Change
`src/ecwam/snonlin_swrt.F90` (new)	`INISNONLIN_SWRT` reads the SWRT file and checks it against the spectral grid. `SNONLIN_SWRT` takes the arguments of `SNONLIN` and adds S_{nl} to `SL` and its diagonal term to `FLD`.
`src/ecwam/yowstat.F90`, `mpuserin.F90`	New switch `INLSCHEME` in namelist `NALINE`, 0 for DIA (default) and 1 for SWRT.
`src/ecwam/userin.F90`	Writes the scheme to `wam.log` and stops for other values.
`src/ecwam/initmdl.F90`	Calls `INISNONLIN_SWRT` when `INLSCHEME = 1`.
`src/ecwam/implsch.F90`, `wdfluxes.F90`	Call `SNONLIN_SWRT` instead of `SNONLIN` when `INLSCHEME = 1`.
`src/ecwam/CMakeLists.txt`	Adds the new file to the build.
`share/ecwam/scripts/ecwam_run_model.sh`	Passes `isnonlin`, `inlscheme` and `swrt_operator` of the yml settings to the namelist and the environment.

Four properties of `SNONLIN_SWRT` matter for ecWAM.

- **Vectorization.** The innermost loop runs over the grid points of a block (NPROMA). The arithmetic is in single precision, and subnormal numbers are flushed to zero inside the routine.
- **Directions.** The anticlockwise direction index of the SWRT file is mapped onto the clockwise index of ecWAM.
- **Depth.** The SWRT files hold the deep-water interactions, and the shallow-water factor of `SNONLIN` (`ISNONLIN`) scales them as it scales DIA.
- **Diagonal term.** $\partial S_{nl}/\partial F$ follows Xnl and enters the implicit integration of ecWAM.

2. Installation

The branch can be cloned directly.

```
git clone -b swrt https://github.com/Jialunx/ecwam
```

An existing clone of ecWAM at commit `e1779dc0` takes the same changes from the patch.

```
git checkout e1779dc0aec1210ab6f5339a2582fbb7b6b18e6b
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/ecwam_swrt_e1779dc0.patch
git apply ecwam_swrt_e1779dc0.patch
```

ecWAM then builds as before and needs no new library. The SWRT files are in the folder `swrt/files` of the branch and go to any directory the model can read.

```
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/swrt_ec4x_K31L13.ww3op
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/wrt_table_K807.ww3op
```

SWRT file	Configurations	Offsets	Use
`swrt_ec4x_K31L13.ww3op`	31	13	SWRT, with the smallest training error at four times the S_{nl} cost of DIA
`wrt_table_K807.ww3op`	807	74	WRT, all configurations with coefficient 1, for reference runs

The files are built for the spectral grid of the ECMWF operational configuration, with 36 directions and the 36 source-term frequencies 0.0345 Hz × 1.1^{m-1} (fr1 = 4.177248E-02 at ifre1 = 3). Other spectral grids need new files, which the SWRT scripts generate offline from Xnl.

3. Running

Setting	Where	Value
`INLSCHEME`	namelist `NALINE`	1 for SWRT, 0 for DIA (default)
`WAM_SWRT_OPERATOR`	environment	path of the SWRT file
`ISNONLIN`	namelist `NALINE`	shallow-water scaling 0, 1 or 2, as for DIA
`WAM_SWRT_DIAG`	environment	`0` switches the diagonal term off (optional)

The standalone scripts read the same settings from three lines of the yml file.

```
physics:
    inlscheme: 1
    isnonlin: 0
    swrt_operator: /abs/path/swrt_ec4x_K31L13.ww3op
```

The log file `wam.log` names the scheme, the SWRT file, and its numbers of configurations and offsets. The model stops with a message when the variable is unset, the file is missing, or the file does not match the spectral grid.

4. Test results

4.1 Synthetic spectra

The error ε of the direction-integrated S_{nl} (van Vledder 2006, Eq. 58) is computed against WRT on 300 synthetic spectra that were not used in the selection, half of them single-peaked and half bimodal.

Method	Median ε (%)	95th percentile of ε (%)
SWRT with K = 31	8	18
DIA of ecWAM	125	190

4.2 Global test

The global test extends the regression case `etopo1_oper_an_fc_048` of ecWAM to 24 h on the 36 × 29 spectral grid, from 12 UTC 31 December 2022 to 12 UTC 1 January 2023. It starts from rest with the ECMWF operational 10-m winds and sea ice, the physics of iphys = 1 and time steps of 900 s. The errors ε_{Hs} and ε_{Tz} are the relative differences from the run with WRT, summed over the hourly outputs from t = 1 h and over the sea points where H_s of WRT is at least 0.1 m.

Method	ε_{Hs} (%)	ε_{Tz} (%)	Mean H_s at 24 h (m)
WRT	–	–	1.20
SWRT with K = 31	0.7	0.8	1.19
DIA	15	9	1.35

SWRT keeps the significant wave height within 1 % of WRT at every hour and nearly everywhere, while DIA gives 15 % higher values (Figs. 1 and 2).

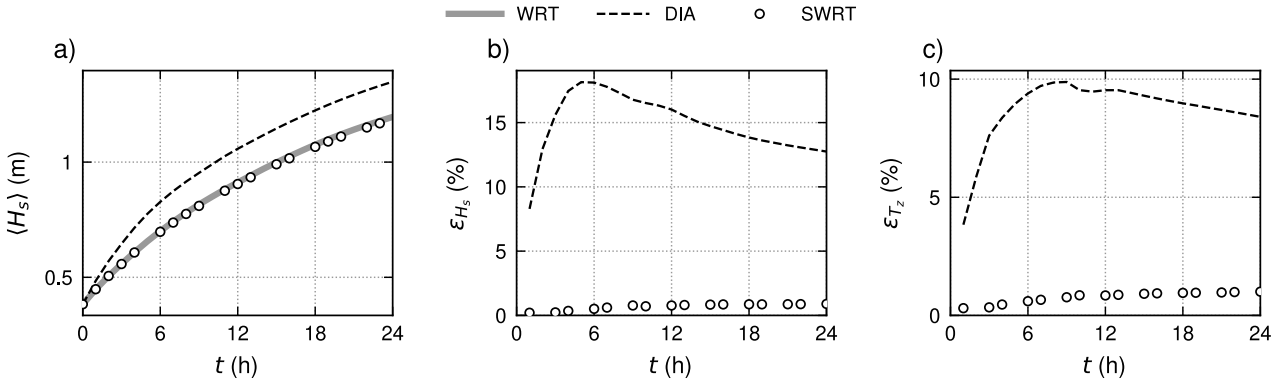


Fig. 1. Global statistics of the 24-h ecWAM test with S_{nl} from WRT (thick gray line), DIA (dashed) and SWRT with K = 31 (circles). (a) Mean H_s over the sea points; (b) ε_{Hs} and (c) ε_{Tz} at each hour.

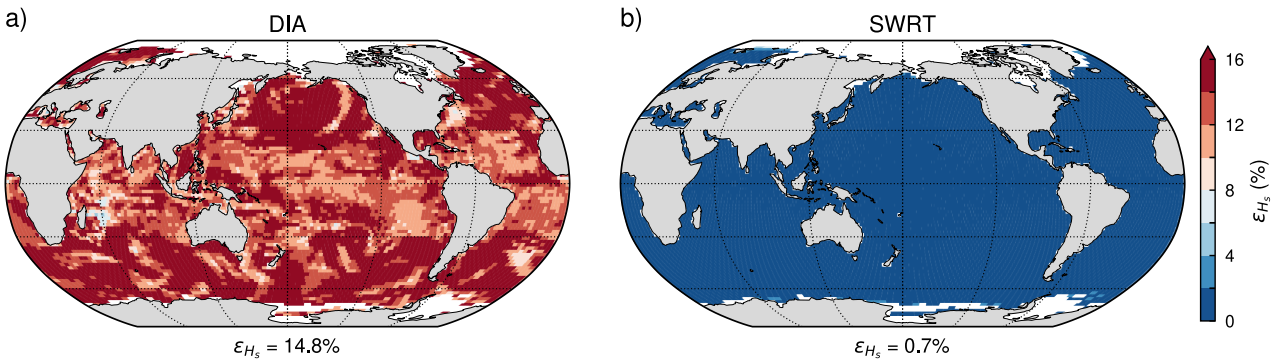


Fig. 2. ε_{Hs} at each sea point over the 24-h ecWAM test for (a) DIA and (b) SWRT with K = 31, with ε_{Hs} over all points under each map. Seas without a value are white, mostly the ice-covered polar seas.

4.3 Cost

The cost is measured with DR_HOOK in the optimized build of the 24-h runs, on one of four MPI tasks, and is given relative to DIA.

Method	S_{nl}	Time step	Run
SWRT with K = 31	4.0	1.6	1.4
WRT	68	14	11

S_{nl} takes about a fifth of a time step with DIA, half with SWRT and nearly all of it with WRT (Fig. 3).

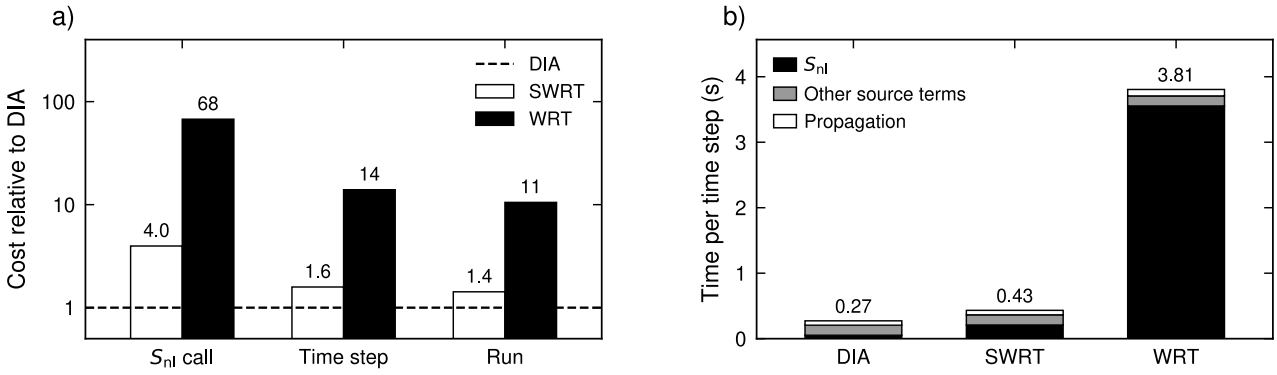


Fig. 3. Cost of S_{nl} in the 24-h ecWAM test. (a) Cost of SWRT with K = 31 (white) and WRT (black) relative to DIA (dashed line) for one S_{nl} call, one time step and the run; (b) time per time step, split into S_{nl}, the other source terms and propagation.